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Signal Processing - HW 5

7. IIR Filters

$$y[n] = 0.5x[n] + 0.5x[n-1] + 0.8y[n-1]$$

10 first samples:

a) $x[n] = \delta[n]$ (1 at $n=0$, 0 elsewhere) $\rightarrow$ $x[n]$

$$x[n] = \delta[n] = \begin{cases} 1 & n=0 \\ 0 & n \neq 0 \end{cases}$$

initial rest:

$$x[n] = 0, y[n] = 0 \quad \text{for } n < 0$$

$$n=0$$

$$y[0] = y[0]$$

$$= 0.5x[0] + 0.5x[-1] + 0.8y[-1]$$

$$0.5 \cdot 1 + 0.5 \cdot 0 + 0.8 \cdot 0$$

$$= 0.5$$

$$n = 1$$

$$\begin{aligned} h[1] &= y[1] \\ &= 0.5x[1] + 0.5x[0] + 0.8y[0] \\ &= 0.5 \cdot 0 + 0.5 \cdot 1 + 0.8 \cdot 0.5 \\ &= 0.9 \end{aligned}$$

$$n = 2$$

$$\begin{aligned} h[2] &= y[2] \\ &= 0.5x[2] + 0.5x[1] + 0.8y[1] \\ &= 0 + 0 + 0.8 \cdot 0.9 \\ &= 0.72 \end{aligned}$$

$$n = 3$$

$$h[3] = 0.8 h[2] = 0.8 \cdot 0.72 = 0.576$$

$$n = 4$$

$$h[4] = 0.8 h[3] = 0.8 \cdot 0.576 = 0.4608$$

$$n=5$$

$$h[5] = 0.8 h[4] = 0.8 \cdot 0.4608 = 0.36864$$

$$n=6$$

$$h[6] = 0.8 h[5] = 0.8 \cdot 0.36864 = 0.294912$$

$$n=7$$

$$h[7] = 0.8 h[6] = 0.8 \cdot 0.294912 = 0.2359296$$

$$n=8$$

$$h[8] = 0.8 h[7] = 0.8 \cdot 0.2359296 = 0.18874368$$

$$n=9$$

$$h[9] = 0.8 h[8] = 0.8 \cdot 0.18874368 = 0.150994944$$

* first 10 samples of $h[n]$ are :

$$h[0] = 0.5$$

$$h[1] = 0.9$$

$$h[2] = 0.72$$

$$h[3] = 0.576$$

$$h[4] = 0.4608$$

$$h[5] = 0.36864$$

$$h[6] = 0.294912$$

$$h[7] = 0.2359296$$

$$h[8] = 0.18874368$$

$$h[9] = 0.150994944$$

b) $x = [1, -1, 1, 1, -1, -1, 0, 0, 0, 0]$ for $n = 0, \dots, 9$
 with $x[n] = 0$ for $n < 0$ and $n > 9$

* $n = 0$

$$x[0] = 1, x[-1] = 0, y[-1] = 0$$

$$y[0] = 0.5x[0] + 0.5x[-1] + 0.8y[-1]$$

$$0.5 \cdot 1 + 0.5 \cdot 0 + 0.8 \cdot 0$$

$$= 0.5$$

* $n = 1$

$$x[1] = -1, x[0] = 1, y[0] = 0.5$$

$$y[1] = 0.5x[1] + 0.5x[0] + 0.8y[0]$$

$$0.5(-1) + 0.5(1) + 0.8(0.5)$$

$$= -0.5 + 0.5 + 0.4 = 0.4$$

* $n = 2$

$$x[2] = 1, x[1] = -1, y[1] = 0.4$$

$$y[2] = 0.5x[2] + 0.5x[1] + 0.8y[1]$$

$$= 0.5(1) + 0.5(-1) + 0.8(0.4)$$

$$= 0.5 - 0.5 + 0.32$$

$$= 0.32$$

* $n = 3$

$$x[3] = 1, x[2] = 1, y[2] = 0.32$$

$$y[3] = 0.5x[3] + 0.5x[2] + 0.8y[2]$$

$$= 0.5 + 0.5 + 0.256$$

$$= 1.256$$

$$n = 4$$

$$x[4] = -1, x[3] = 1, y[3] = 1.256$$

$$y[4] = 0.5x[4] + 0.5x[3] + 0.8y[3]$$

$$= 0.5 + 0.5 + 1.0048$$

$$= 1.0048$$

$$n = 5$$

$$x[5] = -1, x[4] = -1, y[4] = 1.0048$$

$$y[5] = 0.5x[5] + 0.5x[4] + 0.8y[4]$$

$$= -0.5 - 0.5 + 0.80384$$

$$= -0.19616$$

$$n = 6$$

$$x[6] = 0, x[5] = -1, y[5] \approx -0.19616$$

$$y[6] = 0.5x[6] + 0.5x[5] + 0.8y[5]$$

$$= -0.656928$$

$$n=7 \quad x[7]=0, x[6]=0, y[6] \approx -0.656978$$

$$y[7] = 0.5x[7] + 0.5x[6] + 0.8y[6]$$

$$= -0.5255421$$

$$n=8 \quad x[8]=0, x[7]=0, y[7] \approx -0.5255421$$

$$y[8] = 0.8y[7]$$

$$= -0.42043372$$

$$n=9 \quad x[9]=0, x[8]=0, y[8] \approx -0.42043372$$

$$y[9] = 0.8y[8]$$

$$= -0.336347136$$

2. System Diagram & Transfer Function

$$y[n] = \frac{1}{2}x[n] + \frac{1}{3}y[n-1] + \frac{1}{4}y[n-2]$$

$$y[n] - \frac{1}{3}y[n-1] - \frac{1}{4}y[n-2] = \frac{1}{2}x[n]$$

$$Y(z) - \frac{1}{3}z^{-1}Y(z) - \frac{1}{4}z^{-2}Y(z) = \frac{1}{2}X(z)$$

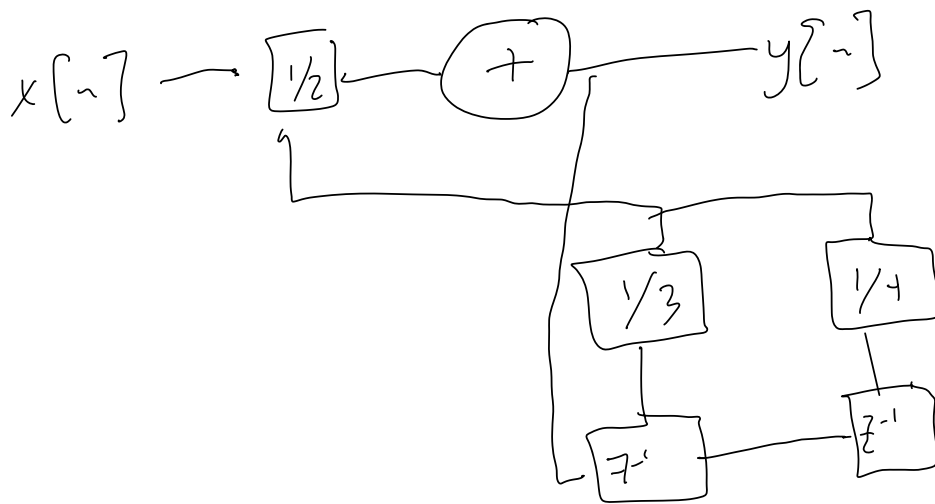
$$Y(z) \left(1 - \frac{1}{3}z^{-1} - \frac{1}{4}z^{-2}\right) = \frac{1}{2}X(z)$$

So:

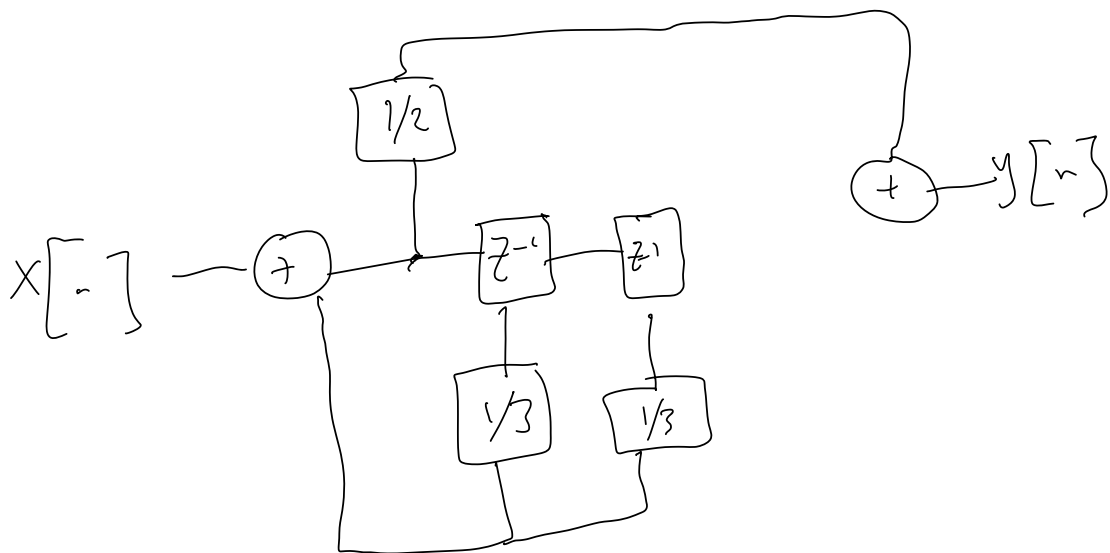
$$H(z) = \frac{Y(z)}{X(z)} = \frac{\frac{1}{2}}{1 - \frac{1}{3}z^{-1} - \frac{1}{4}z^{-2}}$$

Direct form 1:

$$y[n] = 0.5x[n] + \frac{1}{3}y[n-1] + \frac{1}{4}y[n-2]$$



direct form II:



$$\begin{aligned}
 6) \quad y[n] &= x[n] + \frac{1}{2} x[n-1] - \frac{4}{5} y[n-1] \\
 &\quad - \frac{1}{2} y[n-2]
 \end{aligned}$$

match so:

$$y[n] = b_0 x[n] + b_1 x[n-1] + b_2 x[n-2] - a_1 y[n-1] - a_2 y[n-2]$$

So:

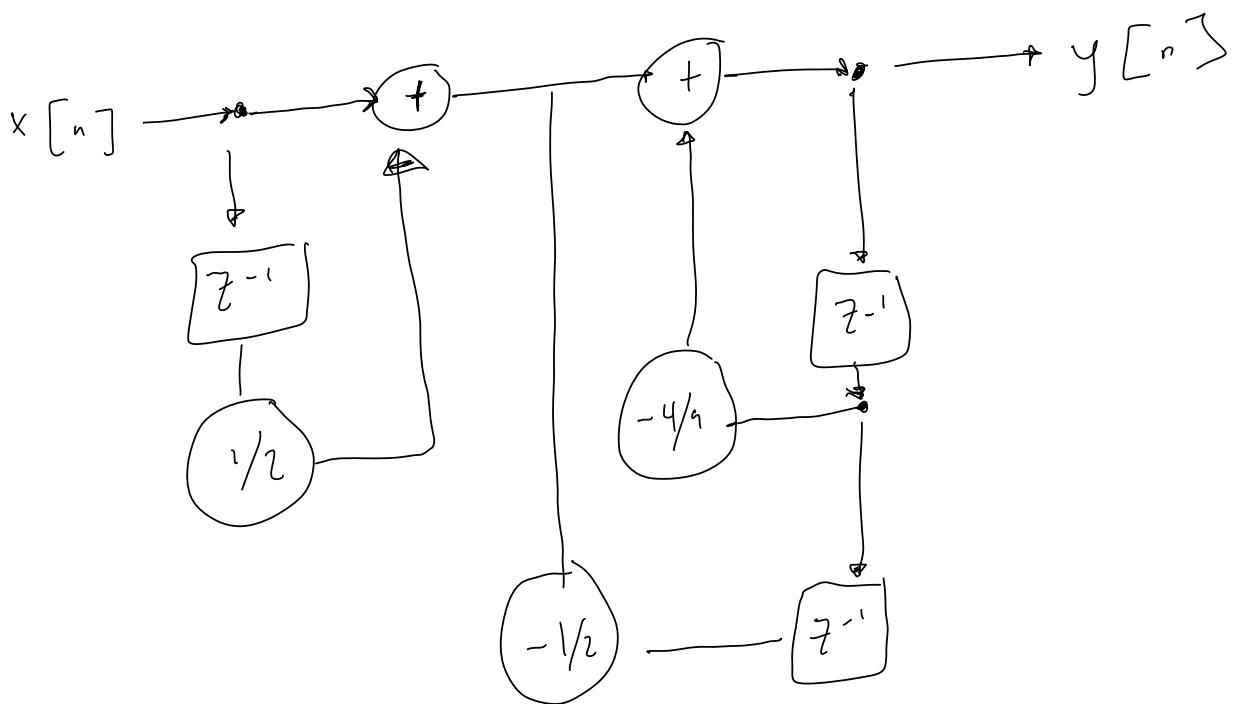
$$b_0 = 1, b_1 = \frac{1}{2}, b_2 = 0$$

$$-a_1 = -\frac{4}{5} \Rightarrow a_1 = \frac{4}{5}$$

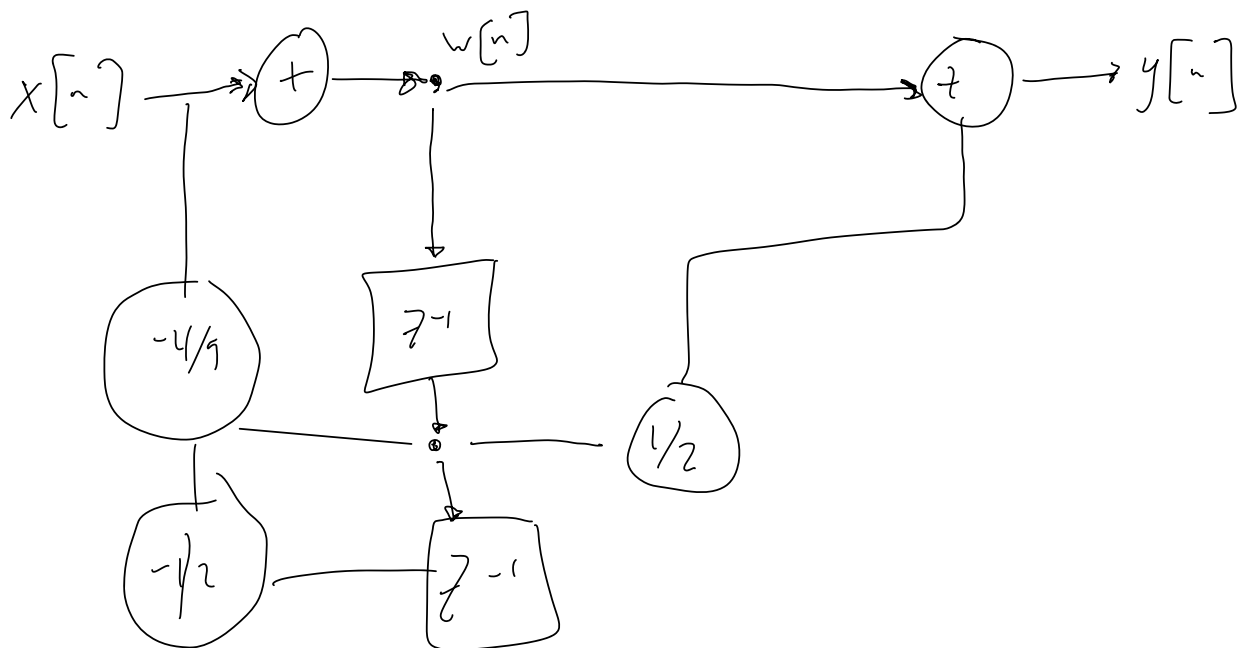
$$-a_2 = -\frac{1}{2} \Rightarrow a_2 = \frac{1}{2}$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1 + \frac{1}{2}z^{-1}}{1 + \frac{4}{5}z^{-1} + \frac{1}{2}z^{-2}}$$

direct form 1



direct form II



3. Partial Fraction Expansion

a) $H_1(z) = \frac{-z}{1 - \frac{4}{3}z^{-1} + \frac{1}{3}z^{-2}}$

factor denominator: $(1 - z^{-1})(1 - \frac{1}{3}z^{-1})$

Zeros: numerator is $-z \Rightarrow$ double zero at $z=0$
 $z_1 = 0$ (mult 2)

poles:

$$p_1 = 1, \quad p_2 = \frac{1}{3}, \quad m_1 = m_2 = 1$$

Stability: pole at $|z| = 1 \rightarrow$ not BIBO-stable

$$H_2(z) = \frac{4 - \frac{7}{2}z^{-1}}{1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}}$$

factor denominator:

$$1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2} = (1 - z^{-1})\left(1 - \frac{1}{2}z^{-1}\right)$$

factor numerator:

$$4 - \frac{7}{2}z^{-1} = z^{-1}\left(4z - \frac{7}{2}\right) \Rightarrow z\left(4z - \frac{7}{2}\right) = 0$$

* zeros:

$$z_1 = 0, \quad z_2 = \frac{7}{8}, \quad \text{both mult. 1}$$

* poles:

$$p_1 = 1, p_2 = \frac{1}{2}, m_1 = m_2 = 1$$

* since $|p_1|, |p_2| < 1$ and $|z| = 1 \rightarrow$ not BIBO-stable

$$H_3(z) = \frac{1}{(1 - \frac{1}{3}z^{-1})(1 - z^{-1} + \frac{1}{4}z^{-2})}$$

Second factor:

$$1 - z^{-1} + \frac{1}{4}z^{-2} = \left(1 - \frac{1}{2}z^{-1}\right)^2$$

So:

$$H_3(z) = \frac{1}{(1 - \frac{1}{3}z^{-1})(1 - \frac{1}{2}z^{-1})^2}$$

* zeros :

numerator is $1 \rightarrow$ triple zero at $z=0$

$$z_1 = 0 \text{ (mult. 3)}$$

* poles :

$$p_1 = \frac{1}{3}, m_1 = 1; \quad p_2 = \frac{1}{2}, m_2 = 2$$

* stability : all poles strictly inside unit circle $\Rightarrow$ stable

$$b) \sum_{i=0}^{Np-1} \sum_{j=0}^{m_i-1} \frac{r_{ij}}{(1 - p_i z^{-1})^{m_i-j}}$$

$$H_1(z)$$

with the factorization above :

$$H_1(z) = \frac{-2}{(1-z^{-1}) \left(1 - \frac{1}{3} z^{-1}\right)}$$

partial fractions:

$$H_1(z) = \frac{-3}{1-z^{-1}} + \frac{1}{1-\frac{1}{3}z^{-1}}$$

$$H_2(z)$$

$$H_2(z) = \frac{4 - \frac{7}{2}z^{-1}}{(1-z^{-1})(1-\frac{1}{2}z^{-1})}$$

partial fractions:

$$H_2(z) = \frac{1}{1-z^{-1}} + \frac{3}{1-\frac{1}{2}z^{-1}}$$

$$H_3(z)$$

$$H_3(z) = \frac{1}{(1-\frac{1}{3}z^{-1})(1-\frac{1}{2}z^{-1})^2}$$

$$H_3 = \frac{4}{1 - \frac{1}{3}z^{-1}} - \frac{6}{1 - \frac{1}{2}z^{-1}} + \frac{3}{(1 - \frac{1}{2}z^{-1})^2}$$



4)

$$z^{-1} \left\{ \frac{1}{1 - az^{-1}} \right\} = a^n u[n],$$

$$z^{-1} \left\{ \frac{1}{(1 - az^{-1})^2} \right\} = (n+1)a^n u[n]$$

a) $H_1(z) \rightarrow \text{fun } 3 \rightarrow H_1(z) = \frac{-2}{1 - \frac{4}{3}z^{-1} + \frac{1}{3}z^{-2}} = \frac{-3}{1-z^{-1}} + \frac{1}{1 - \frac{1}{3}z^{-1}}$

So :

$$h_1[n] = \left(\left(\frac{1}{3} \right)^n - 3 \right) u[n]$$

$$b) H_2(z)$$

$$H_2(z) = \frac{4 - \frac{7}{2}z^{-1}}{1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}} = \frac{1}{1 - z^{-1}} + \frac{3}{1 - \frac{1}{2}z^{-1}}$$

So:

$$h_2[n] = \left(1 + 3\left(\frac{1}{2}\right)^n\right) u[n]$$

$$c) H_3(z)$$

$$H_3(z) = \frac{1}{\left(1 - \frac{1}{3}z^{-1}\right)\left(1 - z^{-1} + \frac{1}{4}z^{-2}\right)} = \frac{4}{1 - \frac{1}{3}z^{-1}} - \frac{6}{1 - \frac{1}{2}z^{-1}} + \frac{3}{\left(1 - \frac{1}{2}z^{-1}\right)^2}$$

So:

$$h_3[n] = \left(4\left(\frac{1}{3}\right)^n - 6\left(\frac{1}{2}\right)^n + 3(n+1)\left(\frac{1}{2}\right)^n\right) u[n]$$

$$\hookrightarrow h_3[n] = \left(4\left(\frac{1}{3}\right)^n + 3(n-1)\left(\frac{1}{2}\right)^n\right) u[n]$$